3.4

ADULTS

Ophthalmology Infections

CHAPTER 3.4	OPHTHALMOLOGY INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Blepharitis			
Chronic Blepharitis (Including associated with Rosacea) Mild to moderate symptoms	**Symptomatic measures:** Warm compress, lid hygiene, artificial tears	**If failed symptomatic measures, or severe symptoms**: Chloramphenicol 0.5% Eye Drops q6H	• Duration: 2 – 4 weeks (long term for anti-inflammatory effects).
Chronic moderate to severe symptoms (that have inadequate response to topical therapy)	Chloramphenicol Eye Drops 0.5% q4-6H **OR** Doxycycline 100 mg PO q12H for 2 weeks, then	Azithromycin 500 mg PO q24H on for 1 day, then 250 mg PO q24H for 4 days	

CHAPTER 3.4	OPHTHALMOLOGY INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
	100 mg PO q24H		
Meibomian Abscess (Internal Hordeolum) *Staphylococcus aureus*	**Symptomatic measures:** Warm compress, lid hygiene Cloxacillin 500 mg IV/PO q6H	**Symptomatic measures:** Warm compress, lid hygiene Cefazolin 1-2 g IV q8H **OR** Cefalexin 500 mg PO q6H	• Duration: Treat until resolution of abscess. • Beta-lactam allergy or concern regarding MRSA infection: Co-trimoxazole 5-10 mg/kg/day of TMP component.
Stye (External Hordeolum) *Staphylococcus aureus*	**Symptomatic measures:** Warm compress and lid hygiene Systemic antibiotics NOT indicated		• For ophthalmology review to ensure no abscess or superficial cellulitis. If present, to refer to antibiotic regimen in the internal hordeolum section.
Cellulitis			

CHAPTER 3.4	OPHTHALMOLOGY INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Orbital (Post Orbital) *H influenzae* *S pneumoniae* *S anginosus/milleri group* *S aureus* Aerobic Gram-negative Anaerobe If intracranial extension is suspected	Co-amoxiclav 1.2 g IV q8H **Followed by oral step-down:** Co-amoxiclav 625 mg PO q8H Ceftriaxone 2 g IV q12H + Metronidazole 500 mg IV q8H		• Duration: 7 – 10 days. • Surgical drainage is required if there is worsening of signs and symptoms. • For IV to PO conversion once clinical response is seen. • If high risk for MRSA, suggest to add Vancomycin 15 mg/kg (max 1 g) q8-12H. • **Penicillin Allergy**: Ciprofloxacin 750 mg PO q12H + Metronidazole 400 mg PO q8H
Periorbital (Preseptal) **<5 years:** *S aureus* *S pneumoniae* *S anginosus/mill*eri group *Haemophilus influenzae* Type B in unvaccinated	Cloxacillin 1 g PO q6H **OR** Cefalexin 500 mg PO q6H	Co-amoxiclav 625 mg PO q8H* **Penicillin Allergy:** Clindamycin 300 mg PO q6H	• Duration: 5-7 days • *Co-amoxiclav for patients <5 years who did not complete vaccination.

CHAPTER 3.4	OPHTHALMOLOGY INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
>5 years: Likely Staphylococcal			
Dacryocystitis			
Mild	Cloxacillin 1 g PO q6H	Cefalexin 500 mg PO q6H **OR** Co-amoxiclav 625 mg PO q8H	• Aspiration and drainage are recommended. • Duration: 7-10 days or till resolution.
Acute severe *S aureus* *S pyogenes*	Cefazolin 2 g IV q8H **OR** Co-amoxiclav 1.2 g IV q8H		
Conjunctivitis			
Mild	Chloramphenicol 0.5% Eye Drops q6H	Tobramycin 0.3% Dexamethasone 0.1%	• Duration: 5 – 7 days

CHAPTER 3.4	OPHTHALMOLOGY INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
	+ Chloramphenicol 1% Eye Ointment	Eye Drops + Gentamicin 0.3% Eye Drops + Tobramycin 0.3% Dexamethasone 0.1% Eye Ointment	• *Levofloxacin 0.5% Eye Drops is not available in UMMC.
Severe	Tobramycin 0.3% / Dexamethasone 0.1% Eye Drops	Moxifloxacin 0.5% Eye Drops 1 drop q8-12H **OR** Levofloxacin 0.5% Eye Drops* 1 drop q8H	
Chlamydia Conjunctivitis (Acute and Trachoma)			
Inclusion Conjunctivitis *Chlamydia trachomatis*	Azithromycin 1 g PO single dose	Doxycycline 100 mg PO q12H	• Duration of Doxycycline: 7 days.
Trachoma	Azithromycin 1 g PO	Doxycycline 100 mg PO	• Duration of Doxycycline: 21 days.

CHAPTER 3.4	OPHTHALMOLOGY INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
	single dose	q12H	
***Neisseria Gonococcal* Conjunctivitis**			
Neisseria gonorrhoeae	Ceftriaxone 1 g IV/IM single dose ± Azithromycin* 1 g PO single dose	**Penicillin Allergy:** Doxycycline 100 mg PO q12H	• Duration of Doxycycline: 7 days. • *Add Azithromycin if *Chlamydia trachomatis* is suspected. • If *Neisseria meningitidis*: consider chemoprophylaxis or treatment for meningococcal disease. • If ophthalmia neonatorum: to treat mother and sexual partner.
Cornea			
Keratitis **Dendritic Ulcer** Common: HSV (Dendritic ulcer)	Chloramphenicol 1% Eye Ointment LA q8H + Aciclovir 3% Eye Ointment* + Aciclovir 800 mg PO five times daily		• Fluorescein staining for presumptive diagnosis, NAT for rapid confirmation. • Consider long term Aciclovir for recurrent HSV keratitis. • *Aciclovir 3% Eye Ointment is not available in UMMC. • *Use for Varicella ophthalmicus. • Duration: 7-10 days.

CHAPTER 3.4	OPHTHALMOLOGY INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Corneal Ulcer Viral: VZV Bacterial: *S aureus, S pyogenes, S pneumoniae, Enterobacteriaceae, P aeruginosa*	Gentamicin 1.4% Eye Drops + Vancomycin 5% Eye Drops **OR** Moxifloxacin 0.5% Eye Drops		
Contact Lenses *P aeruginosa* Fungi *Mycobacterium acanthamoeba*	Tobramycin 0.3%/ Dexamethasone 0.1% Eye Drops **OR** Gentamicin 0.3% Eye Drops		
Retinitis			
Acute Retinal Necrosis VZV, HSV	Aciclovir 10 mg/kg IV q8H for 7-10 days	Valaciclovir 1g PO q8H for 7-10 days	• Duration: 6 weeks. • For CMV retinitis, refer **Chapter 3.13: Human Immunodeficiency Virus Related Opportunistic Infections**

CHAPTER 3.4	OPHTHALMOLOGY INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
	Followed by: Aciclovir 800 mg PO five times daily for 6 weeks **Secondary prophylaxis:** Aciclovir 400 mg PO q12H for 6-12 months **May consider combination with intravitreal antimicrobial:** Forscarnet* 2.4 mg in 0.1 mL IVT (frequency depending on clinical response)	**Followed by:** Valaciclovir 500 mg PO q12H for 6 weeks **May consider combination with intravitreal antimicrobial:** Ganciclovir 2 mg in 0.1 mL IVT (frequency depending on clinical response)	for management. • *Forscarnet is not available in UMMC.
Progressive Outer Retinal Necrosis (PORN) VZV	Aciclovir 10 mg/kg IV q8H for 14-21 days **Followed by:** Aciclovir 800 mg PO five		• Duration: 6 weeks. • For CMV retinitis, refer **Chapter 3.13: Human Immunodeficiency Virus Related Opportunistic Infections** for management. • For PORN, consider Forscarnet IV if not responding to

CHAPTER 3.4	OPHTHALMOLOGY INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
	times daily for 6 weeks **Secondary prophylaxis:** Aciclovir 400 mg PO q12H for 6-12 months **For rapidly progressing disease, may consider combination with intravitreal antimicrobial:** Forscarnet* 2.4 mg in 0.1 mL IVT		Aciclovir IV. • *Forscarnet is not available in UMMC.
Endophthalmitis			
Empirical (Haematogenous, Endogenous) Streptococci Staphylococci Gram-negative bacilli	**Topical Antimicrobial:** Gentamicin 0.3% or 1.4% Eye Drops + Ceftazidime 5% Eye Drops		• Duration: 4-6 weeks.

CHAPTER 3.4	OPHTHALMOLOGY INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
	OR Vancomycin 5% Eye Drops + Moxifloxacin 0.5% Eye Drops **Systemic Antimicrobial:** Ceftriaxone 2 g IV q24H + Vancomycin 15 mg/kg (max 1 g) q8-12H		
If fungal endophthalmitis is suspected based on ocular structure	**Topical Antimicrobial:** Amphotericin B 0.15% Eye Drops **OR** Voriconazole 1% Eye	**Intravitreal Antimicrobial:** Amphotericin B 5 mcg in 0.1 mL IVT	• Duration: 4-6 weeks. • Please take blood cultures (bacterial and fungal) and vitreous cultures (bacterial and fungal).

CHAPTER 3.4	OPHTHALMOLOGY INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
	Drops **Intravitreal Antimicrobial:** Voriconazole 0.1 mg in 0.1 mL IVT **If systemic infection is suspected, ADD:** Amphotericin B Deoxycholate 1-1.5 mg/kg IV q24H		
Post Ocular Surgery, Post Intravitreal Injection, Post Penetrating Trauma (Exogenous Endophthalmitis)	**Topical Antimicrobial:** Vancomycin 5% Eye Drops + Gentamicin 0.3% or 1.4% Eye Drops + Ceftazidime 5% Eye Drops	**Systemic Antimicrobial:** Ciprofloxacin 200 mg IV q12H + Vancomycin 15 mg/kg (max 1 g) q8-12H	• Vitrectomy and lens removal in chronic post ocular surgery.

CHAPTER 3.4	OPHTHALMOLOGY INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
	Systemic Antimicrobial: Ceftazidime 2 g IV q8H + Vancomycin 15 mg/kg (max 1 g) q8-12H **Intravitreal Antimicrobial:** Vancomycin 1 mg in 0.1 mL IVT + Ceftazidime 2-2.25 mg in 0.1 mL IVT		
Ophthalmic Zoster			
Ophthalmic Zoster	Aciclovir 800 mg PO five times daily	Aciclovir 10 mg/kg IV q8H	• Duration: 7 days.
Ocular Tuberculosis			

CHAPTER 3.4	OPHTHALMOLOGY INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Mycobacterium tuberculosis	**AKURIT-4** (Ethambutol + Isoniazid + Rifampicin + Pyrazinamide) for 2 months **Then,** **AKURIT-2** (Isoniazid + Rifampicin) for 4 months **PLUS** Pyridoxine 20 mg PO q24H		• Positive IGRA has poor predictive value and should not be used to confirm diagnosis of TB. It cannot distinguish between latent and active-TB. • Intravitreal sampling must be sent for mycobacterial cultures, bacterial and fungal cultures and TB PCR. • To do HIV testing. • To refer to the Infectious Diseases team for further management. • Ethambutol may cause optic neuropathy and regular eye reviews are required.
Ocular Syphilis			
Treponema pallidum	Syphilitic uveitis should be treated as Neurosyphilis. Refer to **Chapter 3.11: Genital and Sexually Transmitted Infections** for management.		• To perform HIV testing. • To refer to the Infectious Diseases team for further management.